

**Exercise 2.1.** For each of the following, give three examples of this property. Then, prove that it is true.

- (a) The sum of an even integer and an odd integer is odd;
- (b) The product of two even integers is even;
- (c) The product of two odd integers is odd;
- (d) The product of an even integer and an odd integer is even;
- (e) An even integer squared is an even integer.

**Solution:**

To prove each statement, we use the definitions:

- An **even integer** has the form  $2k$ , where  $k \in \mathbb{Z}$ .
- An **odd integer** has the form  $2k + 1$ , where  $k \in \mathbb{Z}$ .

(a) The sum of an even integer and an odd integer is odd

Three examples

$$2 + 3 = 5$$

$$8 + 7 = 15$$

$$14 + 11 = 25$$

Each result is odd.

**Proof**

Let the even integer be

$$2m$$

and the odd integer be

$$2n + 1$$

where  $m, n \in \mathbb{Z}$ .

Their sum is

$$2m + (2n + 1)$$

$$= 2m + 2n + 1$$

$$= 2(m + n) + 1$$

Since  $m + n$  is an integer,  $2(m + n) + 1$  has the form of an odd integer.

Therefore,

|                                                       |
|-------------------------------------------------------|
| The sum of an even integer and an odd integer is odd. |
|-------------------------------------------------------|

**(b) The product of two even integers is even**

**Three examples**

$$2(4) = 8$$

$$6(10) = 60$$

$$8(12) = 96$$

Each result is even.

**Proof**

Let the two even integers be

$$2m \quad \text{and} \quad 2n$$

Their product is

$$(2m)(2n) = 4mn$$

We can rewrite this as

$$4mn = 2(2mn)$$

Since  $2mn$  is an integer, the product has the form  $2k$ .

Therefore,

|                                           |
|-------------------------------------------|
| The product of two even integers is even. |
|-------------------------------------------|

**(c) The product of two odd integers is odd**

**Three examples**

$$3(5) = 15$$

$$7(9) = 63$$

$$11(13) = 143$$

Each result is odd.

**Proof**

Let the two odd integers be

$$2m + 1 \quad \text{and} \quad 2n + 1$$

Their product is

$$(2m + 1)(2n + 1)$$

Expanding:

$$= 4mn + 2m + 2n + 1$$

Factor out 2:

$$= 2(2mn + m + n) + 1$$

Since  $2mn + m + n$  is an integer, this expression has the form

$$2k + 1$$

which is odd.

Therefore,

|                                         |
|-----------------------------------------|
| The product of two odd integers is odd. |
|-----------------------------------------|

(d) The product of an even integer and an odd integer is even

Three examples

$$2(3) = 6$$

$$8(5) = 40$$

$$14(7) = 98$$

Each result is even.

**Proof**

Let the even integer be

$$2m$$

and the odd integer be

$$2n + 1$$

Their product is

$$(2m)(2n + 1)$$

Factor out 2:

$$= 2[m(2n + 1)]$$

Since  $m(2n + 1)$  is an integer, the product has the form  $2k$ .

Therefore,

The product of an even integer and an odd integer is even.

**(e) An even integer squared is an even integer**

Three examples

$$2^2 = 4$$

$$6^2 = 36$$

$$10^2 = 100$$

Each result is even.

**Proof**

Let an even integer be

$$2m$$

where  $m \in \mathbb{Z}$ .

Squaring it gives

$$\begin{aligned}(2m)^2 \\ = 4m^2\end{aligned}$$

Rewrite this as

$$= 2(2m^2)$$

Since  $2m^2$  is an integer,  $4m^2$  has the form  $2k$ , which is even.

Therefore,

|                                        |
|----------------------------------------|
| The square of an even integer is even. |
|----------------------------------------|

**Main idea**

For these proofs, the key is to start with the definition of even and odd:

|             |
|-------------|
| Even = $2k$ |
|-------------|

|                |
|----------------|
| Odd = $2k + 1$ |
|----------------|

Then use algebra until the result can be written in one of those forms.